13	**B**	Since the initial transmitted intensity is zero, the polarisation axis is perpendicular to the transmission axis of the first filter. The transmission axis of the second filter is 40° to that of the first filter. Intensity I_1 transmitted through the second filter: $I_1 = I_0 \cos^2\theta = I_0 \cos^2(90° - 40°) = I_0 \cos^2 50°$ $I_1 = 0.413 I_0$ New intensity I_2 transmitted through the first filter: $I_2 = I_1 \cos^2\theta = I_1 \cos^2(40° - 0°) = I_1 \cos^2 40°$ $I_1 = 0.413 I_0 \times 0.587 = 0.24 I_0$
14	**D**	$I = kA^2$